1. If the area A of a circle is increasing at the rate of $\frac{8\pi}{9}$ cm^2/min, then the rate of change of the radius of the circle when $A = \frac{\pi}{9}$ cm^2 is

(a) $\frac{4}{3}$ cm/min

(b) $\frac{2}{9}$ cm/min

(c) $\frac{1}{3}$ cm/min

(d) $\frac{2}{3}$ cm/min

(e) $\frac{1}{6}$ cm/min

Given $\frac{dA}{dt} = \frac{8\pi}{9}$ cm^2/min,

Find $\left.\frac{dr}{dt}\right|_{A=\frac{\pi}{9}\text{ cm}^2}$

$A = \pi r^2$, $\frac{\pi}{9} = \pi r^2 \Rightarrow \boxed{r = \frac{1}{3}}$

Now $\frac{dA}{dt} = 2\pi r \frac{dr}{dt} \Rightarrow$

$\frac{8\pi}{9} = 2\pi\left(\frac{1}{3}\right)\frac{dr}{dt} \Rightarrow$

$\left.\frac{dr}{dt}\right|_{\substack{r=\frac{1}{3} \\ A=\frac{\pi}{9}}} = \frac{4}{3}$ cm/min

2. If $F(y) = \left(y^{-2} - 3y^{-3}\right)\left(y^{-2} + 3y^{-3}\right)$, then $F'(1) =$

(a) 40

(b) -45

(c) -50

(d) 45

(e) 50

$F(y) = y^{-4} - 9y^{-6} \Rightarrow$

$F'(y) = -4y^{-5} + 54y^{-7}$

$\Rightarrow F'(1) = -4 + 54 = 50$

3. $\cosh\left(\frac{1}{3}\ln x\right) + \sinh\left(\frac{1}{3}\ln x\right) =$

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(a) $e^{\sqrt[3]{x}}$

(b) $2\sqrt[3]{x}$

(c) $\sqrt[3]{x}$

(d) $e^{\sqrt[3]{x}} + e^{-\sqrt[3]{x}}$

(e) $\sqrt[3]{x} + \frac{1}{\sqrt[3]{x}}$

4. $\frac{d}{dx}\left(\frac{e^{x+1} - e^x}{e^x}\right) = \frac{d}{dx}\left(\frac{e^{x+1}}{e^x} - \frac{e^x}{e^x}\right)$

$= \frac{d}{dx}(e - 1) = 0$

(a) e^x

(b) 0

(c) e^{x+1}

(d) e

(e) 1

5. The position of a particle is given by the equation $s = f(t) = 2t^3 - 27t^2 + 108t$ where t is measured in seconds and s in meters. If the particle is moving in the negative direction on the largest time interval (α, β), then $5\alpha - 2\beta =$

(a) 2
(b) 10
(c) 1
~~(d)~~ 3
(e) 9

$v = \frac{ds}{dt} = 6t^2 - 54t + 108$

$= 6(t^2 - 9t + 18)$

$= 6(t-3)(t-6)$

v: + on $(-\infty, 3)$, − on $(3, 6)$, + on $(6, \infty)$

$\Rightarrow v < 0$ on $(3,6)$

$\Rightarrow \alpha = 3, \beta = 6 \Rightarrow$

$5\alpha - 2\beta = 15 - 12 = 3$

6. The y-intercept of the normal line to the graph of $y = \frac{1-3x}{1-6x}$ at $x = \frac{1}{3}$ is

(a) $\left(0, \frac{1}{21}\right)$
(b) $\left(0, -\frac{1}{21}\right)$
(c) $\left(0, -\frac{1}{18}\right)$
(d) $\left(0, \frac{1}{12}\right)$
~~(e)~~ $\left(0, \frac{1}{9}\right)$

$y' = \frac{(1-6x)(-3) - (1-3x)(-6)}{(1-6x)^2}$

$= \frac{-3 + 18x + 6 - 18x}{(1-6x)^2} = \frac{3}{(1-6x)^2}$

$y'(\frac{1}{3}) = \frac{3}{(1-2)^2} = 3$

The slope of the normal line $= -\frac{1}{3}$

Point $\left(\frac{1}{3}, y(\frac{1}{3})\right) \equiv (\frac{1}{3}, 0)$

$\Rightarrow$ Eqn of normal line $y = -\frac{1}{3}(x - \frac{1}{3})$

$x = 0 \Rightarrow y = \frac{1}{9} \Rightarrow$ y-intercept $(0, \frac{1}{9})$

7. If $y = \frac{1}{3}\tan^{-1}\left(\frac{x+1}{3}\right)$, then $\frac{dy}{dx} =$

(a) $9(x^2+2x+10)^{-1}$

(b) $x(x^2+2x+10)^{-1}$

(c) $(x^2+2x+10)^{-1}$

(d) $3(x+1)(x^2+2x+10)^{-1}$

(e) $\frac{1}{3}(x+1)(x^2+2x+10)^{-1}$

$\frac{d}{dx}\tan^{-1}u = \frac{\frac{du}{dx}}{1+u^2}$

$\Rightarrow y' = \frac{1}{3}\cdot\frac{\frac{1}{3}}{1+\frac{(x+1)^2}{9}}$

$= \frac{1}{9+(x+1)^2} = \frac{1}{9+x^2+2x+1}$

$= [x^2+2x+10]^{-1}$

8. The equation of the tangent line to the curve $y = \frac{\sec x}{x^2}$ at $x = \pi$, is

(a) $y = -\frac{3}{\pi^2}x + \frac{2}{\pi^2}$

(b) $y = -\frac{4}{\pi^3}x + \frac{3}{\pi^2}$

(c) $y = \frac{2}{\pi^3}x + \frac{1}{\pi^2}$

(d) $y = \frac{1}{\pi^3}x - \frac{2}{\pi^2}$

(e) $y = \frac{2}{\pi^3}x - \frac{3}{\pi^2}$

at $x=\pi \Rightarrow y = \frac{\sec\pi}{\pi^2} = \frac{-1}{\pi^2}$

Point of tangency is $(\pi, -\frac{1}{\pi^2})$

$y' = \frac{x^2\sec x\tan x - 2x\sec x}{x^4}$

$y'(\pi) = \frac{\pi^2(-1)(0) - 2\pi(-1)}{\pi^4}$

$= \frac{2}{\pi^3}$ = slope of the tangent line at $(\pi, \frac{-1}{\pi^2})$

$\Rightarrow$ The required equation

$y + \frac{1}{\pi^2} = \frac{2}{\pi^3}(x-\pi) = \frac{2}{\pi^3}x - \frac{2}{\pi^2}$

$\Rightarrow y = \frac{2}{\pi^3}x - \frac{3}{\pi^2}$

9. If $3x^2 - 4xy + y^2 = 15$, then $\frac{dy}{dx} =$

(a) $\frac{3x-2y}{2x-y}$

(b) $\frac{3x+2y}{2x-y}$

(c) $\frac{3}{2+y}$

(d) $\frac{3x-6y}{2-y}$

(e) $\frac{2x-3y}{x+2y}$

$\frac{d}{dx}(3x^2-4xy+y^2) = \frac{d}{dx}(15) \Rightarrow$

$6x - 4(xy' + y) + 2yy' = 0 \Rightarrow$

$y'(-4x+2y) = -6x + 4y \Rightarrow$

$y' = \frac{-6x+4y}{-4x+2y}$

$= \frac{3x-2y}{2x-y}$

10. If $y = \frac{x^{2/3}(x-1)^{1/3}}{x+2}$, then $y'(2) =$ [Hint: Use logarithmic differentiation].

(a) $\frac{11\sqrt[3]{4}}{24}$

(b) $\frac{9\sqrt[3]{2}}{8}$

(c) $\frac{5\sqrt[3]{4}}{48}$

(d) $\frac{13\sqrt[3]{4}}{48}$

(e) $\frac{7\sqrt[3]{2}}{48}$

$\ln y = \frac{2}{3}\ln x + \frac{1}{3}\ln(x-1) - \ln(x+2)$

$\frac{y'}{y} = \frac{2}{3x} + \frac{1}{3(x-1)} - \frac{1}{x+2}$

$\Rightarrow y' = \frac{x^{2/3}(x-1)^{1/3}}{(x+2)}\left[\frac{2}{3x} + \frac{1}{3(x-1)} - \frac{1}{x+2}\right]$

$\Rightarrow y'(2) = \frac{(2^{2/3})(1)}{4}\left[\frac{1}{3} + \frac{1}{3} - \frac{1}{4}\right]$

$\Rightarrow y'(2) = \frac{\sqrt[3]{4}}{4}\left[\frac{2}{3} - \frac{1}{4}\right] = \frac{\sqrt[3]{4}}{4}\left(\frac{5}{12}\right) = \frac{5\sqrt[3]{4}}{48}$

11. If $f(x) = \sin^2 x$, then $f^{(5)}(x) =$

(a) $\cos^2 x$

(b) $16(\cos^2 x - \sin^2 x)$

(c) $32 \sin x \cos x$

(d) $8 \sin x \cos x$

(e) $16 \sin^2 x \cos x$

$f'(x) = 2\sin x \cos x = \sin 2x$

$f''(x) = 2\cos 2x$

$f'''(x) = -4\sin 2x$

$f^{(4)}(x) = -8\cos 2x$

$f^{(5)}(x) = 16\sin 2x$

$= 32 \sin x \cos x$

12. If $L(x) = (f \circ g \circ h)(x)$, where $h(1) = 2$, $g(2) = 3$, $h'(1) = 4$, $g'(2) = 5$, and $f'(3) = 6$, then $L'(1) =$

(a) 720

(b) 240

(c) -50

(d) 50

(e) 120

$L(x) = f(g(h(x)))$

Apply the chain Rule :

$L'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$

Now at $x=1 \Rightarrow g(h(1)) = g(2) = 3$

$\Rightarrow L'(1) = f'(g(h(1))) \cdot g'(h(1)) \cdot h'(1)$

$= f'(3) \cdot g'(2) \cdot 4$

$= (6) \cdot (5) \cdot (4) = 120$

13. If α and β are constants such that the function

$$f(x) = \begin{cases} \alpha x + \beta, & x < 0 \\ 2\sin x + 3\cos x, & x \geq 0 \end{cases}$$

is differentiable, then $f(-5) =$

(a) 9
(b) −7
(c) 3
(d) −5
(e) −8

f(x) is differentiable $\Rightarrow$ f is continuous

$\Rightarrow \lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = f(0) \Rightarrow$

$\lim_{x\to 0^-} (\alpha x + \beta) = \lim_{x\to 0^+} (3\sin x + 3\cos x)$

$\Rightarrow \boxed{\beta = 3}$. Also $\lim_{x\to 0^-} f'(x) = \lim_{x\to 0^+} f'(x)$

$\Rightarrow \lim_{x\to 0^-} \alpha = \lim_{x\to 0^+} (2\cos x - 3\sin x) \Rightarrow \boxed{\alpha = 2} \Rightarrow$

$f(-5) = (2)(-5) + (3) = -7$

14. If $f(x) = \cot\left(\frac{\pi}{4}\sqrt{\cot 2x}\right)$, then $f'\left(\frac{\pi}{8}\right) =$

(a) $\frac{\pi}{4}$
(b) π
(c) $\frac{1}{2}$
(d) 4
(e) $\frac{\pi}{2}$

$f'(x) = -\csc^2\left(\frac{\pi}{4}\sqrt{\cot 2x}\right) \cdot \frac{d}{dx}\left[\frac{\pi}{4}(\cot 2x)^{1/2}\right]$

$= -\csc^2\left(\frac{\pi}{4}\sqrt{\cot 2x}\right) \cdot \frac{\pi}{4} \cdot \frac{1}{2}(\cot 2x)^{-1/2}(-2\csc^2 2x)$

$\Rightarrow f'\left(\frac{\pi}{8}\right) = -\csc^2\left(\frac{\pi}{4}\right) \cdot \frac{\pi}{4} \cdot \frac{1}{2}(-2(2))$

$= (-2)\left(\frac{\pi}{4}\right)\left(\frac{-4}{2}\right) = \pi$

15. If $f(x) = \sqrt{1+x^2}$ where $x > 0$, then $f(\sinh x) + f'(\operatorname{csch} x) =$

(a) $\cosh x + \coth x$

(b) $\cosh x + \operatorname{sech} x$

(c) $2\cosh x$

(d) $\cosh x + \sinh x$

(e) $2\cosh x + 1$

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16. If $Ax^2 + By^2 = C$, where $A, B,$ and C are nonzero constants, then $y'' =$

(a) $-\frac{AC}{B^2y^3}$

(b) $-\frac{ABC}{y^3}$

(c) $-\frac{ACx}{By^2}$

(d) $-\frac{AB^2}{Cy^3}$

(e) $-\frac{AC}{B^4y^3}$

$\frac{d}{dx}(Ax^2 + By^2) = \frac{d}{dx}(C) \Rightarrow$

$2Ax + 2Byy' = 0 \Rightarrow$

$y' = -\frac{Ax}{By} \Rightarrow$

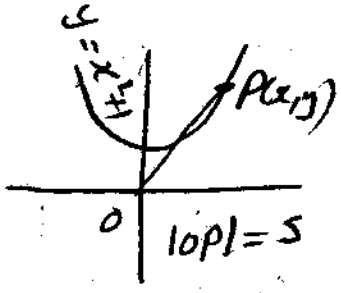
$y'' = \left(-\frac{A}{B}\right)\left(\frac{y - xy'}{y^2}\right) \Rightarrow$

$y'' = -\frac{A}{By^2}\left[y + \frac{Ax^2}{By}\right] \Rightarrow$

$y'' = -\frac{A}{By^2}\,\frac{By^2 + Ax^2}{By} = -\frac{A}{B^2y^3}(C)$

$= \frac{-AC}{B^2y^3}$

17. A particle is moving along the curve $y = x^2 + 1$. As the particle passes through the point $(1,2)$, its y-coordinate increases at a rate of 10 cm/s. At this instant, the distance from the particle to the origin is changing at a rate of



Distance $= S = \sqrt{x^2+y^2} \Rightarrow$

(a) 5 cm/s

(b) $6\sqrt{5}$ cm/s $\quad \frac{dS}{dt} = \frac{1}{2}(x^2+y^2)^{-1/2}\left(2x\frac{dx}{dt} + 2y\frac{dy}{dt}\right)$

(c) $2\sqrt{5}$ cm/s

(d) $5\sqrt{5}$ cm/s $\quad$ But $\frac{dy}{dt} = 2\frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{1}{2}\frac{dy}{dt} \Rightarrow$

(e) 20 cm/s $\quad$ at $(1,2)$, $\frac{dy}{dt} = 10$ and $\frac{dx}{dt} = 5 \Rightarrow$

$$\left.\frac{dS}{dt}\right|_{(1,2)} = \frac{1}{2\sqrt{5}}\left((2)(1)(5) + (2)(2)(10)\right) = \frac{50}{2\sqrt{5}} = \frac{25\sqrt{5}}{5}$$

$$= 5\sqrt{5} \text{ cm/sec}.$$

18. If $f(x) = (\sin^{-1} x)^{x+\frac{1}{2}}$, then $f'\left(\frac{1}{2}\right) =$

(a) $\frac{\pi}{6}\ln\frac{\pi}{6} + \sqrt{3}$

(b) $\frac{\pi}{3}\ln\frac{\pi}{3} + \frac{3\sqrt{3}}{2}$

(c) $\frac{\pi}{6}\ln\frac{\pi}{6} + \frac{2\sqrt{3}}{3}$

(d) $\frac{\pi}{6}\ln\frac{\pi}{6} + \frac{\pi\sqrt{3}}{18}$

(e) $\frac{\pi}{3}\ln\frac{\pi}{3} + \frac{\pi\sqrt{3}}{3}$

$\ln f(x) = (x+\frac{1}{2})\ln\sin^{-1}x \Rightarrow$

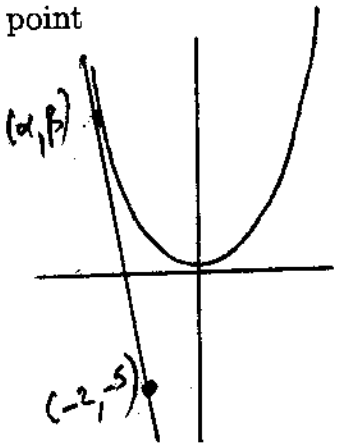
$$\frac{f'(x)}{f(x)} = \left(\ln\sin^{-1}x\right) + \frac{x+\frac{1}{2}}{(\sin^{-1}x)\sqrt{1-x^2}}$$

$$\Rightarrow f'(x) = (\sin^{-1}x)^{x+1/2}\left[\ln\sin^{-1}x + \frac{x+\frac{1}{2}}{(\sin^{-1}x)\sqrt{1-x^2}}\right]$$

$$\Rightarrow f'(\tfrac{1}{2}) = (\tfrac{\pi}{6})\left[\ln\tfrac{\pi}{6} + \frac{1}{\frac{\pi}{6}\cdot\frac{\sqrt{3}}{2}}\right]$$

$$\Rightarrow f'(\tfrac{1}{2}) = \frac{\pi}{6}\ln\frac{\pi}{6} + \frac{2}{\sqrt{3}} = \frac{\pi}{6}\ln\frac{\pi}{6} + \frac{2\sqrt{3}}{3}.$$

19. Let T be the tangent line to the parabola $y = x^2$ at the point $P(\alpha, \beta)$. If $\alpha < 0$ and T passes through the point $(-2, -5)$, then the slope of T is equal to



$y = x^2 \Rightarrow \boxed{\beta = \alpha^2}$

(a) -10

$y' = 2x \Rightarrow$

(b) 2

slope of $T = 2\alpha = \frac{\beta+5}{\alpha+2}$

(c) -25

(d) -15

$\Rightarrow 2\alpha^2 + 4\alpha = \alpha^2 + 5 \Rightarrow$

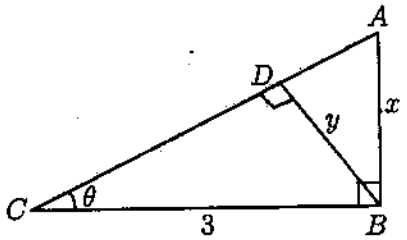
(e) 5

$\alpha^2 + 4\alpha - 5 = 0 = (\alpha+5)(\alpha-1)$

$\Rightarrow \alpha = -5, \ \alpha \neq 1 \Rightarrow (\alpha, \beta) = (-5, 25)$

slope of $T = \frac{\beta+5}{\alpha+2} = \frac{25+5}{-5+2} = \frac{30}{-3} = -10$

20. The given figure shows a right triangle ABC at B and BD is perpendicular to AC. If the lengths of AB, BD, and BC are, respectively, x, y, and 3, then $\lim_{\theta \to 0^+} \frac{x}{y}$



(a) is equal to $\frac{1}{3}$

(b) does not exist

(c) is equal to 3

(d) is equal to 0

(e) is equal to 1

$\tan\theta = \frac{x}{3}, \ \sin\theta = \frac{y}{3}$

$\Rightarrow \frac{x}{y} = \frac{\tan\theta}{\sin\theta} \Rightarrow$

$\lim_{\theta \to 0^+} \frac{x}{y} = \lim_{\theta \to 0^+} \frac{\tan\theta}{\sin\theta} = \lim_{\theta \to 0^+} \left(\frac{\tan\theta}{\theta}\right)\left(\frac{\theta}{\sin\theta}\right)$

$= 1$